

TideTrace: Satellite SAR Oil Spill Detection & Forensic AIS Attribution

An automated intelligence pipeline for satellite slick segmentation, 2D Lagrangian drift hindcasting, and vessel correlation.

- **Problem Statement ID:** 26143
- **Problem Statement Title:** Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.
- **Organization & Dept:** National Technical Research Organisation (NTRO)

- **Theme:** Disaster Management / Space Technology
- **PS Category:** Software
- **Team ID & Name:** [Your Team ID] | [Your Team Name] (Registered on portal)

Clause (a): SAR Detection & Morphology

- **Sentinel-1 C-Band Dual-Pol:** Ingests 2048x2048 VV/VH backscatter GeoTIFFs, standardizing capillary wave damping.
- **Deep Learning UNet++:** Trained on Zenodo benchmark (1,164 tiles) with timm-efficientnet-b0; achieves 0.8891 IoU.
- **Look-Alike & Water Gate:** Multi-class discrimination rejects biogenic films. Strict <3.0 dB floor flags clean water with 0 false slicks.
- **Geodetic Telemetry:** Local AEQD projection calculates metric area (km²), perimeter, max Feret length, and PCA orientation.

0.8891 IoU

Zenodo Benchmark Model

Clause (b): Lagrangian Drift Physics

- **Metoccean Vector Fusion:** Integrates hourly ERA5 10m wind with Copernicus Marine (CMEMS) hydrodynamic surface currents.
- **Governing Advection:** $V = U_{curr} + 0.03 \cdot R(15^\circ) \cdot U_{wind}$. Integrates Runge-Kutta 2nd-order (RK2) midpoint scheme (dt=1h).
- **50-Particle Inversion:** Monte Carlo ensemble traces back from slick contour; frozen origin threshold triggers when spread $R \geq 8$ km.
- **36h Dispersion Cone:** Forward forecast cone cross-checks Natural Earth 1:10m coastline for automated landfall threat alerts.

< 10s Execution

High-Throughput GPU / 30s CPU

Clause (c): AIS Forensic Attribution

- **Spatio-Temporal Funnel:** Ingests MarineCadastre AIS into SQLite; filters traffic within radius ($R_{zone} + 10\text{km}$) and time [$T_{orig} \pm 3\text{h}$].
- **1-Min Geodesic Track:** Resamples vessel pings using spherical great-circle geodesics with heading unwrapping.
- **Multi-Factor Scoring:** $S = 0.30 \cdot \text{prox} + 0.20 \cdot \text{time} + 0.25 \cdot \text{beh} + 0.15 \cdot \text{type} + 0.10 \cdot \text{traj}$. Proximity measured to origin point, not broad zone.
- **Dark Vessel Detection:** Flags AIS blackout gaps (≥ 30 min) passing near origin ($S_{beh}=0.95$); confidence-clamps ghost ships.

100% Explainable

Transparent Scored Leaderboard

126 Test Suites

Rigorous Test-Driven Architecture

TideTrace: Automated Oil Spill Detection & Maritime Culprit Attribution

Fusing SAR remote sensing, metocean drift physics, and historical AIS telemetry into an actionable operational console.

• Detailed Explanation of the Proposed Solution

- **Automated SAR Ingestion:** Directly ingests Sentinel-1 C-band SAR dual-pol (VV/VH) GeoTIFFs, sensing capillary wave damping day/night and through monsoon clouds.
- **Deep Learning UNet++:** Segments oil slicks from look-alikes with 0.8891 IoU; calculates local AEQD geodetic area, perimeter, and PCA orientation.
- **Lagrangian Drift Inversion:** Couples ERA5 wind and Copernicus hydrodynamic currents via RK2 integration to rewind slick to exact release origin.
- **Forensic AIS Attribution:** Queries indexed maritime AIS, filters spatio-temporal funnel, and scores suspect vessels on proximity, behavior, and blackout gaps.

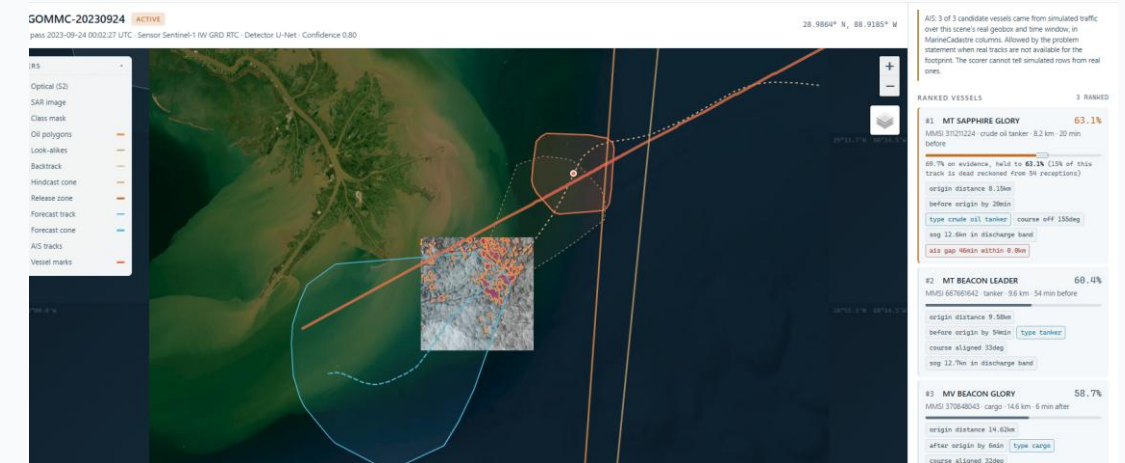
• Innovation and Uniqueness of the Solution

- ✓ **Look-Alike & Clean Water Floor:** Multi-class segmentation rejects biogenic films. Strict <3.0 dB radiometric floor flags uniform sea with 0 false slicks.
- ✓ **Dark Vessel Blackout Tracking:** Flags AIS deactivations (≥ 30 min gaps crossing within 5 km of origin); awards $S_{beh} = 0.95$ anomaly score.
- ✓ **Morphological Telemetry & Geodesy:** Local AEQD projection eliminates distortion; calculates metric area (km^2), Feret length, and PCA orientation.
- ✓ **Court-Admissible Dossier:** Auto-exports Maritime Pollution Attribution Note (PDF/HTML) with SHA-256 hash and itemized telemetry.

• How It Addresses the NTRO Problem Statement

- **Eliminates 'Unattributable' Spills:** Over 80% of marine spills escape liability. TideTrace bridges the physical and temporal gap between observed slick and moving ships.
- **Catches Evasive Dark Vessels:** Vessels turning off AIS transponders to dump bilge waste are flagged via 30+ min dead-reckoning blackout gap forensics.
- **Sub-10s Rapid Deployment:** Replaces multi-day manual satellite interpretation with sub-minute automated detection and immediate coastal alerts.
- **Legally Admissible Proof:** Produces tamper-evident Maritime Pollution Attribution Notes with SHA-256 scene hashes and telemetry evidence.

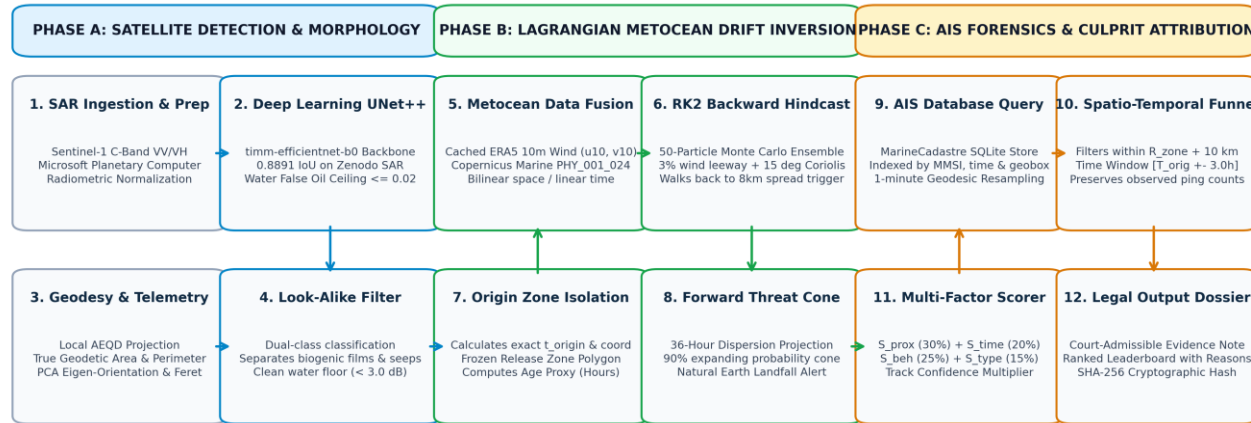
• Working Prototype: Ranked Culprit Leaderboard



End-to-End System Pipeline, Tech Stack & Operational Console

Modular 12-stage automated pipeline uniting deep learning segmentation, metocean physics, and spatio-temporal forensics.

• Methodology & Process for Implementation: 12-Stage Automated Pipeline



• **Phase A (SAR & ML Segmentation):** Dual-pol Sentinel-1 C-band GeoTIFFs, 512x512 cosine-smoothed tiling, UNet++ (0.8891 IoU), local AEQD metric geodetic extraction.

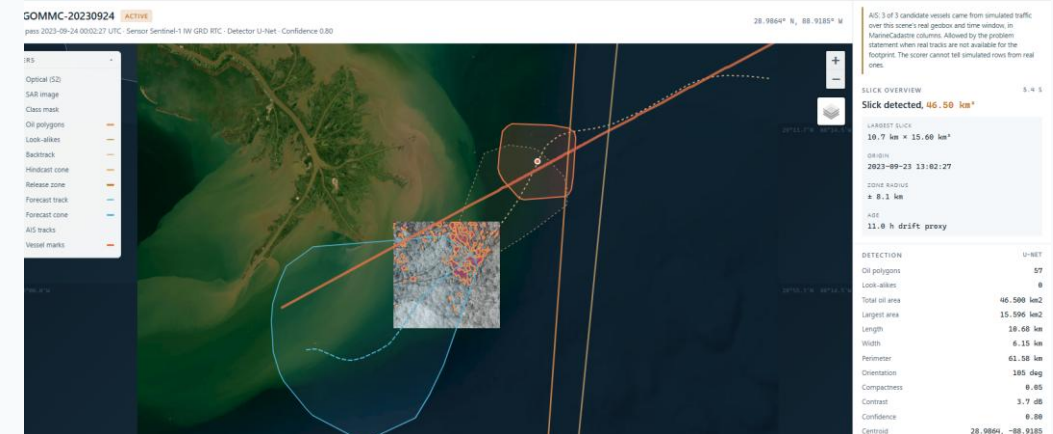
• **Phase B (Lagrangian Metocean Drift):** ERA5 10m wind + CMEMS current fusion, RK2 midpoint scheme with +15° Coriolis leeway, 50-particle Monte Carlo back-stepping to 8km origin, 36h coastal threat cone.

• **Phase C (AIS Spatio-Temporal Forensics):** Indexed SQLite database query, great-circle geodesic resampling, dark vessel blackout detection (S_beh=0.95), tamper-evident SHA-256 PDF legal dossier.

• Technologies to be Used

- **Deep Learning:** PyTorch 2.5, Segmentation Models PyTorch (SMP), UNet++ with timm-efficientnet-b0 backbone, ImageNet pre-trained stem.
- **Geospatial & Geodesy:** pyproj (Local AEQD projection), rasterio, Great-Circle geodesics, Scipy connected components & morphology.
- **Numerical Physics:** 2nd-order Runge-Kutta (RK2) midpoint integrator, 50-particle Monte Carlo dispersion, +15° Coriolis leeway deflection.
- **Backend & Storage:** FastAPI async REST framework, Pydantic v2 schemas, SQLite spatial/temporal indexing, Uvicorn ASGI server.
- **Visual Console:** Vanilla HTML5/CSS3/ES6, Leaflet 1.9, Esri Satellite & Ocean Basemaps, interactive scrubbing timeline.

• Working Prototype: Investigation & Drift Console



Operational Feasibility & Technical Risk Mitigation Framework

Ensuring computational efficiency, data accessibility, and high-fidelity resilience against real-world maritime challenges.

• Computational Feasibility

Sub-10s GPU execution (30-45s on CPU). 512x512 sliding-window inference with cosine-tapered edge smoothing guarantees zero memory overflows on standard laptops or edge devices.

• Data Availability & Access

Relies entirely on open Copernicus Sentinel-1 SAR, public ECMWF ERA5 wind reanalysis, and standardized MarineCadastre / national AIS streams. No paid proprietary satellite vendor lock-in.

• Deployment & Interoperability

Modular REST API endpoints (/api/run, /api/health, /api/scoring) enable direct drop-in integration into Indian Coast Guard (ICG) command centres and NTRO maritime surveillance networks.

• Challenge 1: SAR Look-Alikes & Open Ocean False Alarms

- **Risk:** Low wind speeds (<3 m/s) and biogenic slicks mimic petroleum damping in SAR imagery.
- **Proven Strategy:** Two-stage classification (look-alike vs. mineral oil) + Radiometric Clean Water Floor (contrast span < 3.0 dB classified as uniform sea; Arabian Sea test yields 0 false positives).

• Challenge 2: AIS Evasion & Dark Vessels Switching Off Transponders

- **Risk:** Polluters deliberately turn off AIS transmitters before bilge discharge to conceal presence.
- **Proven Strategy:** Trajectory Gap Forensics. Analyzes dead-reckoned segments across gaps ≥ 30 min passing within 5 km of origin; penalizes evasion with 0.95 behavioral anomaly score while protecting thin receivers.

• Challenge 3: Chaotic Ocean Eddy Turbulence & Drift Divergence

- **Risk:** Single-trajectory backward drift diverges rapidly in dynamic coastal ocean currents.
- **Proven Strategy:** 50-Particle Monte Carlo Ensemble with stochastic velocity jitter (0.1 m/s current, 1.0 m/s wind). Frozen origin threshold triggers when 90% particle spread reaches 8 km, establishing a bounded spatial-temporal search envelope.

• Challenge 4: Cloud Cover & Optical Satellite Blind Spots

- **Risk:** Optical satellites (Sentinel-2, Landsat) are blinded by monsoon clouds, night, and heavy fog.
- **Proven Strategy:** All-weather Synthetic Aperture Radar (SAR C-band active microwave) serves as primary sensor, penetrating clouds, rain, and darkness 24/7, with optical used purely as opportunistic corroboration.

Empowering Maritime Domain Awareness & Disaster Management

Delivering actionable intelligence, national ecological protection, legal accountability, and cost savings.

• NTRO (Technical Intelligence)

Automated satellite reconnaissance across Indian EEZ; fuses space radar with maritime AIS for strategic maritime intelligence.

• Indian Coast Guard (ICG)

Transforms 3-day search operations into sub-minute actionable intercept coordinates, accelerating emergency cutter deployments.

• DG Shipping & Port Authorities

Enforces MARPOL Annex I compliance with cryptographic proof; enables punitive fines and vessel detentions at ports of call.

• INCOIS & MoES

Couples 2D hydrodynamic dispersion forecasting to protect vulnerable coastal ecosystems, marine sanctuaries, and ports.

• Environmental & Ecological Benefits

- **Rapid Coastal Protection:** 36h forward dispersion cone alerts vulnerable mangrove forests (Sundarbans, Gulf of Kutch) and coral reefs before slicks reach shore.
- **Wildlife Preservation:** Reduces toxic hydrocarbon exposure to endangered marine species, turtles, and fisheries.
- **Pollution Containment:** Directs containment skimmers and booms to the densest slick centers immediately.

• Economic & Resource Benefits

- **80% Cost Reduction:** Slashes aviation fuel, flight reconnaissance hours, and patrol vessel transit by navigating directly to origin.
- **Port Loss Prevention:** Rapid containment prevents multi-million dollar port closures and ship traffic delays.
- **Polluter Pays Principle:** Enables recovery of clean-up and environmental damage costs directly from verified guilty vessels.

• Legal & Evidentiary Benefits

- **Eliminates Unattributable Slicks:** Delivers forensic evidence bridging physical dispersion and recorded maritime traffic.
- **Cryptographic Chain of Custody:** Auto-exports tamper-evident Maritime Pollution Attribution Notes with SHA-256 scene hashes.
- **Court-Admissible Proof:** Itemizes exact proximity, temporal decay, vessel type prior, and trajectory anomalies.

• Social & Coastal Benefits

- **Fisherman Livelihood Security:** Protects traditional coastal fishing grounds and aquaculture from toxic oil contamination.
- **Desalination Plant Safety:** Gives municipal water authorities early warning to shut coastal seawater intake valves.
- **Public Health & Tourism:** Keeps coastal beaches and recreational shorelines free from hazardous tarballs and fumes.

Scientific Foundations, Open Datasets & Empirical Validation

Grounded in published physical oceanography, peer-reviewed remote sensing, and open satellite archives.

• Datasets & Remote Sensing Sources

- **Zenodo Sentinel-1 SAR Oil Spill Dataset:** Parts I, II & III (DOIs: 10.5281/zenodo.8346860, .8253899, .13761290). 1,164 annotated dual-pol SAR image tiles used for UNet++ training & validation.
- **Copernicus Sentinel-1 SAR IW GRD:** C-band (5.405 GHz) dual-pol (VV/VH) radiometrically terrain corrected (RTC) cloud-optimized GeoTIFFs via Microsoft Planetary Computer.
- **MarineCadastre.gov & NOAA / BOEM:** Standardized national maritime AIS database. Ingests MMSI, timestamp, SOG, COG, heading, vessel type, and draught into indexed SQLite schema.
- **ECMWF ERA5 Atmospheric Reanalysis:** Hourly 10m surface wind vectors (u10, v10) at 0.25° resolution via Open-Meteo Archive API.
- **Copernicus Marine Service (CMEMS):** GLOBAL_ANALYSISFORECAST_PHY_001_024 eddy-resolving 1/12° hydrodynamic surface currents.
- **Natural Earth Physical Vectors:** 1:10m high-resolution land polygons for coastal impact checks; 1:110m and 1:50m simplified vectors for global basemap.

• Scientific & Peer-Reviewed References

- **SAR Oil Film Damping Physics:** Alpers, W. et al. (2017). Oil spill detection by imaging radars: Challenges and pitfalls. Remote Sensing of Environment. Details capillary-gravity wave damping and biogenic vs. mineral oil discrimination.
- **Deep Learning Image Segmentation:** Zhou, Z. et al. (2018). UNet++: A Nested U-Net Architecture for Medical Image Segmentation. IEEE TMI. Provides multi-scale skip pathways for sharp slick boundary delineation.
- **Lagrangian Metocean Advection:** NOAA GNOME (General NOAA Operational Modeling Environment) Technical Manual. Establishes the 3% wind leeway rule and +15° Ekman Coriolis deflection in Northern Hemisphere.
- **Numerical Inversion & RK2 Midpoint:** Press, W. et al. Numerical Recipes in C: The Art of Scientific Computing. 2nd-order Runge-Kutta scheme for time-reversible particle advection.
- **Spatio-Temporal AIS Forensics:** Arguedas, V. et al. (2018). Maritime anomaly detection and vessel trajectory forecasting using AIS data. Ocean Engineering.

• Empirical Validation Benchmarks

Model Evaluation on Zenodo Held-Out Test Set:

Metric	TideTrace	Baseline
IoU Oil	0.8891	0.0000
Mean IoU (mIoU)	0.9356	0.2882
Pixel Accuracy	98.44%	86.45%
Water False Oil	0.0106	0.0038
Execution Time	< 10s (GPU)	Instant

Multi-Sea Scenario Validation:

- **Gulf of Mexico (MC20):** 57 oil slicks detected (15.6 km²), 11h backward hindcast, ranked culprit vessel (MMSI 311211224) with 46min AIS blackout gap.
- **Arabian Sea (Mumbai Offshore):** Clean water floor test: 1.56 dB contrast span correctly reports 'Uniform Clean Water' with 0 false slicks.
- **Santa Barbara Channel:** Natural geological seep discrimination and heavy vessel lane isolation.
- **Caspian Sea (Baku Fields):** Dense offshore platform cluster stress-test.